AC Circuits

$v_{in}(t) = Vu(t)$ or $V\delta(t)$ or $V\,pulse(t)$

$\downarrow$

$u(t) - 2u(t-T) + 2u(t-2T) - 2u(t-3T)$

for AC circuits:

$$v_{in}(t) = V\cos(\omega_0 t + \phi)\ ;\ \phi = 2\pi f_0$$

RC: $RC\frac{dv_c}{dt} + v_c(t) = v_{in}(t)$

$$I(s) = \frac{V_{in}(s)}{R + 1/sC} \Rightarrow V_c(s) = \frac{I(s)}{sC} = \frac{V_{in}(s)}{sRC+1}$$

$$v_{in}(t) = e^{at}u(t) \Rightarrow V_c(s) = \frac{1}{s-a}\cdot\frac{1}{sRC+1} = \frac{A}{s-a} + \frac{B}{s+1/RC}$$

$$\Rightarrow v_c(t) = Ae^{at} + Be^{-t/RC}$$

↳ natural response

$v_c(0) = V_0 \Rightarrow V_0 = A+B$

If $v_{in}(t)$ is $\exp(at)$, then $v_c(t)$ is linear combination of explatit natural response

Now, recall $v_{in}(t) = Vu(t)$ If $v_c(0) = 0 \Rightarrow v_c(t) = V(1-e^{-t/RC})$

$$v_c(t) = Vu(t) - Ve^{-t/RC}u(t)$$

$Vu(t)$: input; $Ve^{-t/RC}u(t)$: natural response

If $v_c(0) = V_0 \Rightarrow V > V_0 \Rightarrow v_c(t) = V_0 + (V-V_0)(1-e^{-t/RC})u(t)$

$$= Au(t) + Be^{-t/RC}$$